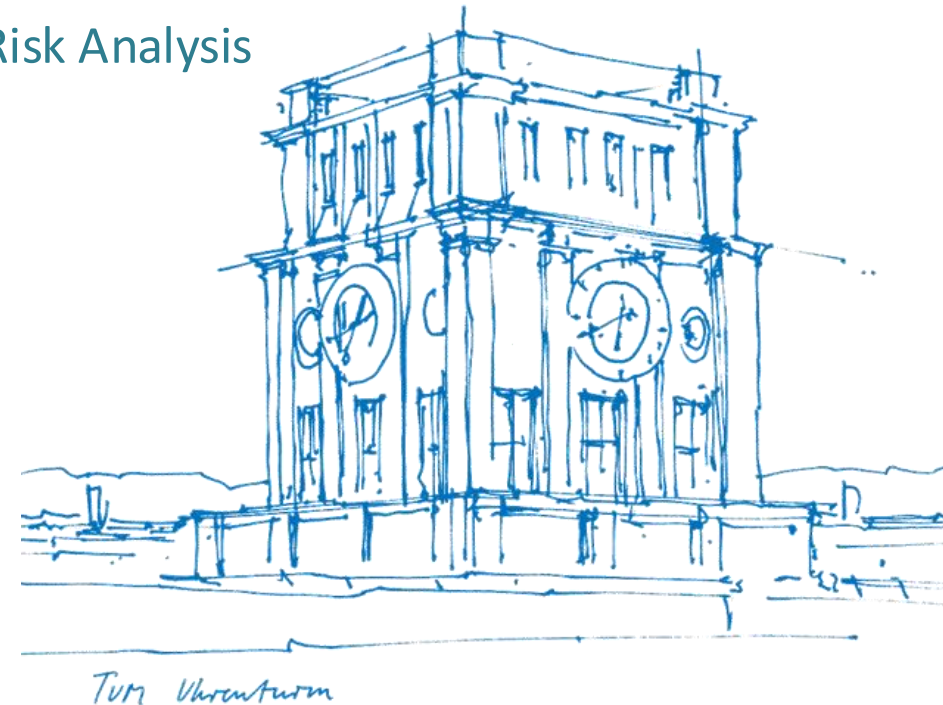


Levels of Building Fire Risk Modeling

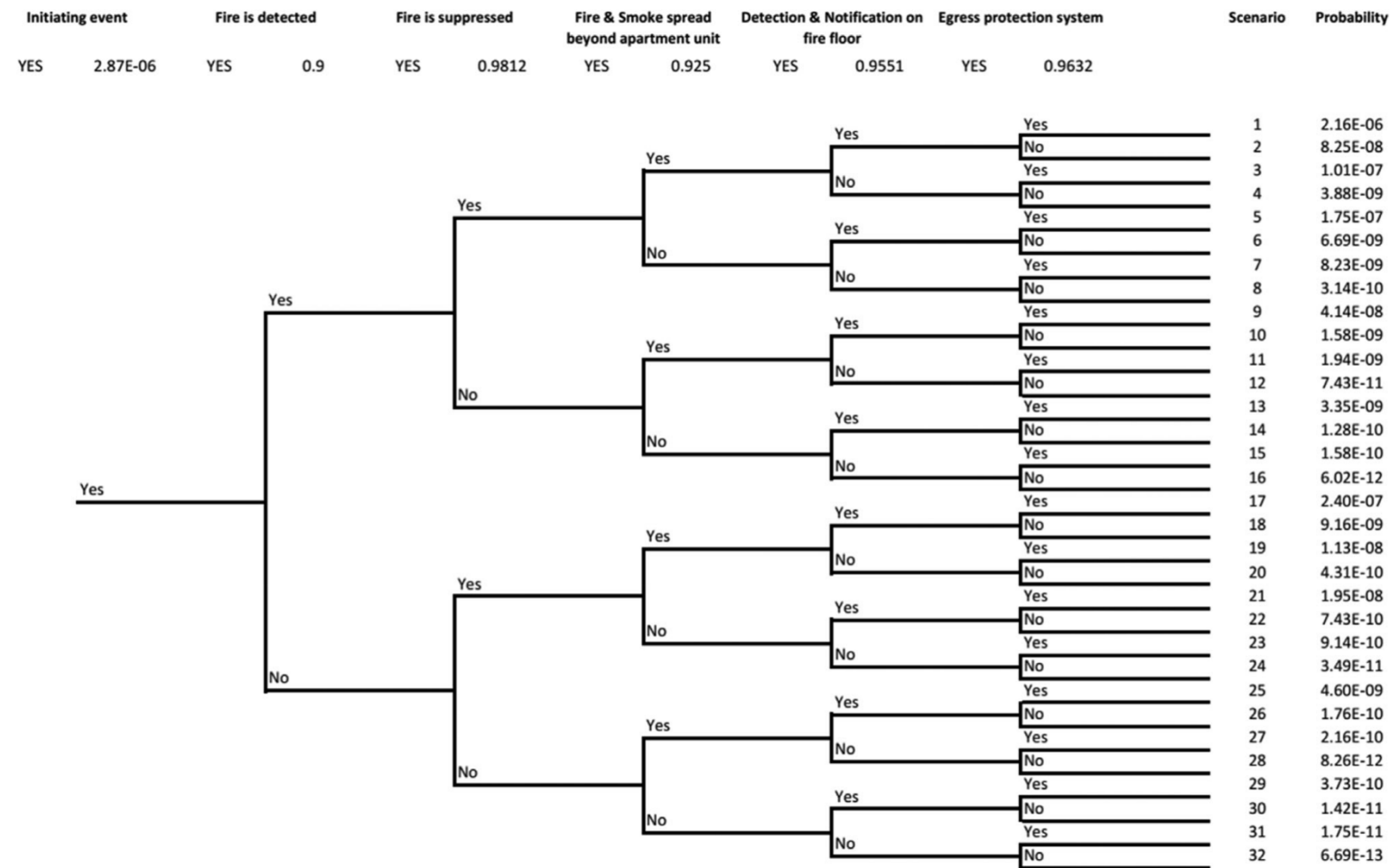
Workflow and Generic Modeling Framework for Community Fire Risk Analysis

Chupei Lin



Event Trees

Probabilistic decomposition of an initiating event into success/failure branches



KEY IDEA

An event tree examines the success and failure of systems following an initiating event (e.g., ignition).

Each branch is assigned a probability; the leaves give the likelihood of every scenario

With the probability of each scenario, simulation tools are used to analyze the consequences.

Strengths

- Transparent decision logic
- Quantifies low-probability tails

Limitation

- Static
- Assumes branch independence and fixed probabilities

Fig. Typical event tree for a fire event in a building. Adapted from Tan et al. (2019).

Bayesian Networks

Directed acyclic graph of variables and conditional dependencies

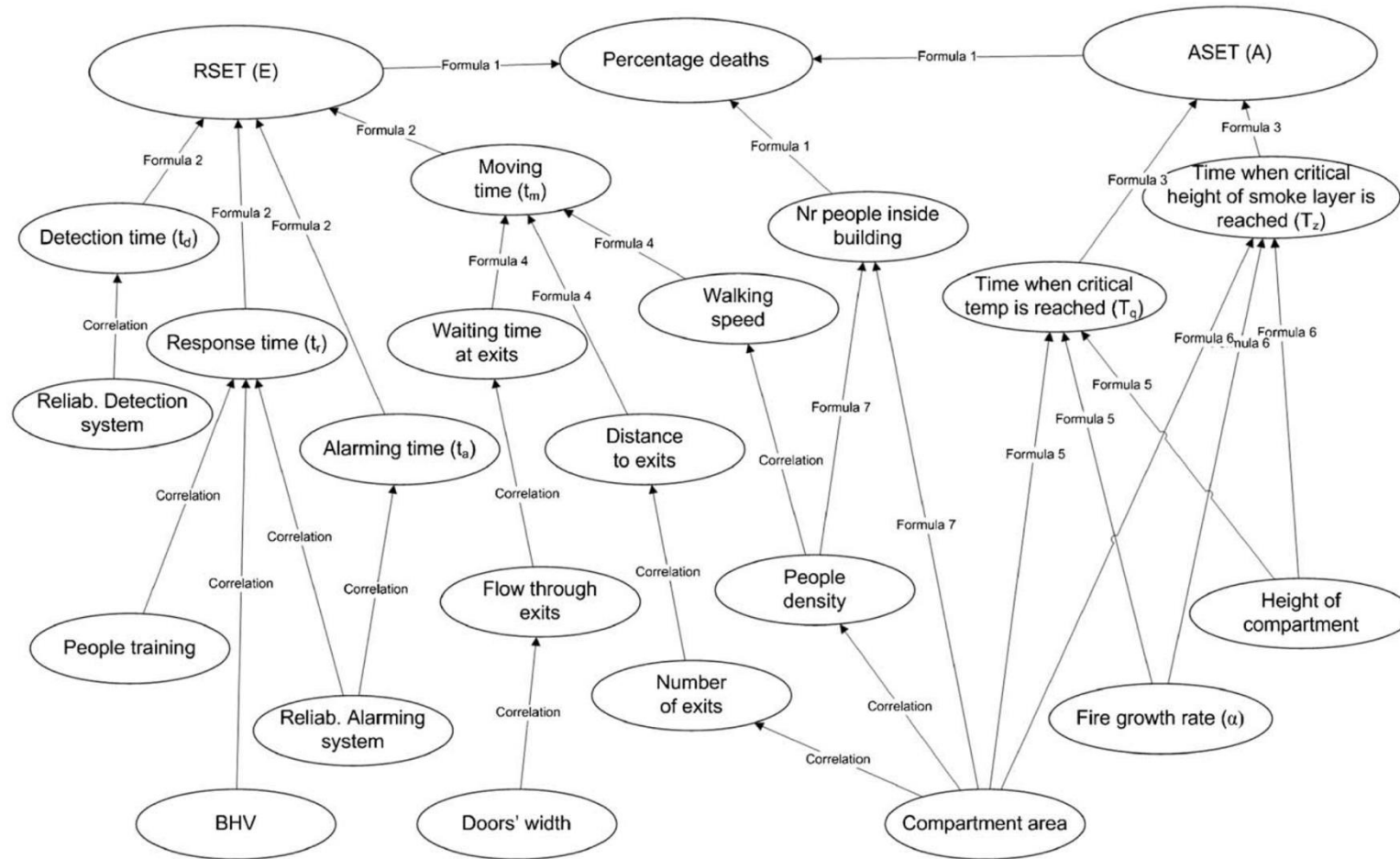


Fig. 2. Model for human fatalities in a fire.

KEY IDEA

Nodes represent random variables;
arcs encode causal relations between
variables.

Computes posterior probabilities of
outcomes given evidence.

Probabilistic, not deterministic.

Strengths

- Handles partial evidence
- Captures correlations between variables
- Updates as new data arrives

Limitation

Requires informed conditional-probability tables.

Traditional Fire-Frequency Models

Volume / floor-area + activation-risk classification

Generalized Barrois model

Tilander & Keski-Rahkonen, 2003

Ignition frequency:

$$f''_m = c_1 A^r + c_2 A^s$$

A is the floor area; c_1 , c_2 , r , s are coefficients, which are determined experimentally from observations for different building categories.

Building categories

- Residential buildings
- Industrial buildings & warehouses
- All other buildings

Matrisk Fire Probability Model (top-down)

Matrisk BSV report, 2025

Annual fire probability:

$$Pr(B|V) = \frac{2}{\pi} \arctan \left(\delta \frac{e^{\alpha} V^{\beta}}{e^{(21 \gamma)}} \right) \left[\frac{1}{a} \right]$$

α , β , γ are determined by activation-risk class; V is the building volume.

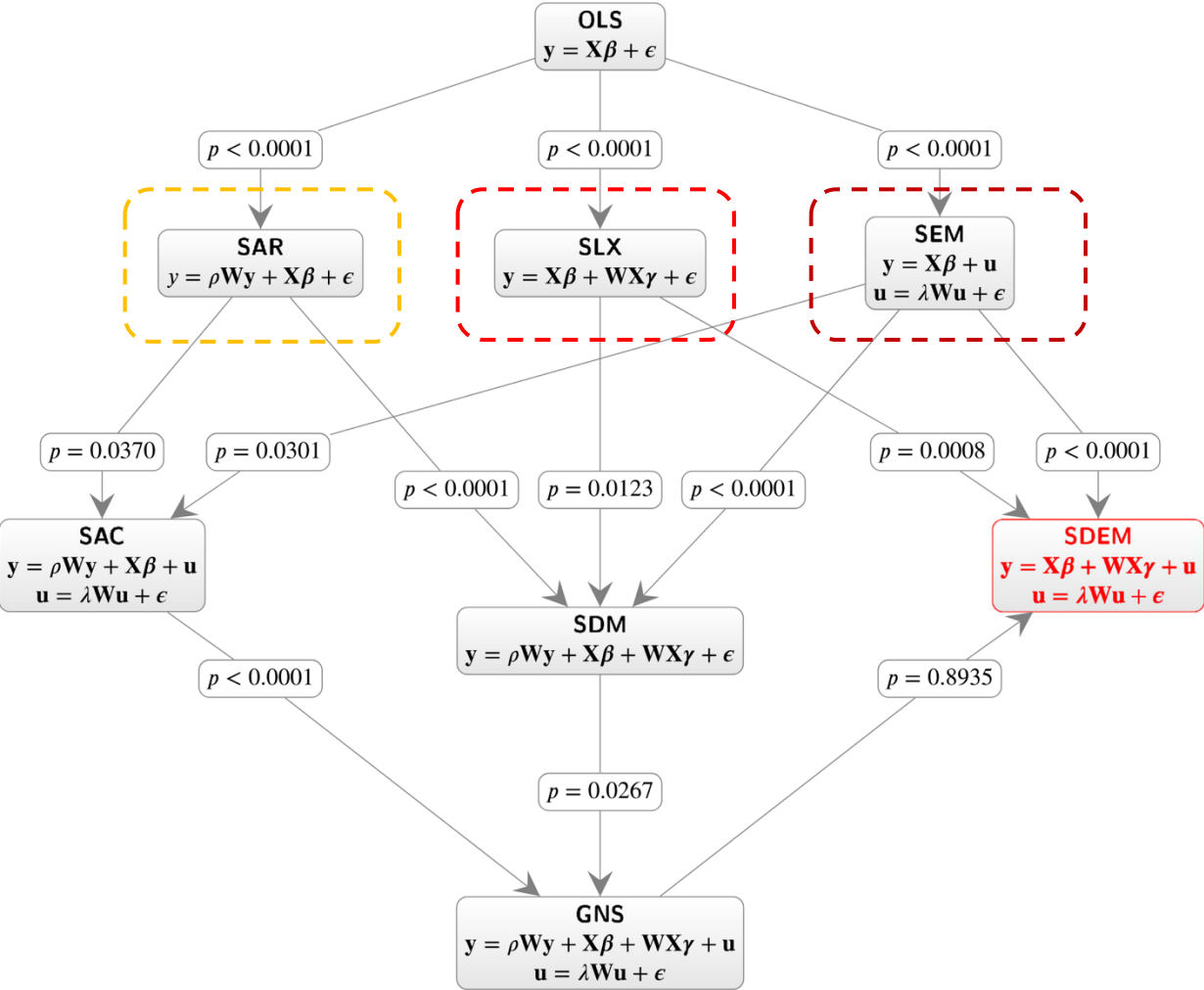
Activation-risk classification (VKG)

Classified based on the purpose code of the Association of Cantonal Building Insurance

No risk	Small	Medium	High
Room without electrical installations, without intended fire and without activating hazards	Retail, Shopping centers, Parking, Library, Hospital rooms	School classrooms, Restaurants (commercial), Industry buildings	Residential use, Hotel rooms, Kitchen use

Spatial Regression Models

OLS extended with three forms of spatial dependence



THREE SPATIAL REGRESSORS

- 1 Spatially lagged dependent variable**
y depends on neighbors' y (ρWy term)
- 2 Spatially lagged explanatory variables**
neighbors' X influence local y (WXγ term)
- 3 Spatially lagged error term**
residuals correlated with neighbors (λWu term)

Description of the explanatory variables.

Variable name	Description
Population density	Number of inhabitants per km ²
Buildings density	Number of buildings per km ²
Degraded buildings density	Number of degraded buildings or in need of repair per km ²
Age of buildings	Average age of buildings (years)
Income	Average income per person (€)
Buying power	Consumer buying power (%)
Education	Secondary school completion rate (%)
Crime	Criminality rate (‰)

The Problem and Research Gap

Fires remain a substantial risk to life, property, and urban infrastructure — risk-reduction decisions operate at two scales

Research Gap · A generic model connecting **occurrence / ignition** and **consequence** for building-level fire-risk analysis.

BUILDING-LEVEL

Consequence analysis

SCALE

Single building

METHODS

Event/fault trees, Bayesian networks

INPUTS

Building & occupant-behavior factors

KEY WORKS

Matellini et al. (2013) · Hanea & Ale (2009) · Tan et al. (T-H-O) (2021)

COMMUNITY-LEVEL

Occurrence prediction

SCALE

Urban area / community / census tracts

METHODS

Spatial regression, machine learning

INPUTS

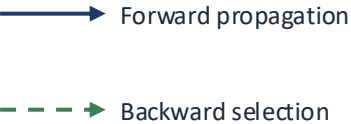
Socio-demographic predictors (income, density, housing age)

KEY WORKS

Buffington et al. (2021) · Bispo et al. (2023) · Hao et al. (2023)

Multi-Scale Fire Risk for Community Risk Management

Forward propagation generates risk; backward selection refines models and data



Training procedure

Prediction procedure

Building-level

Community-level

DATA

Inputs (training set)
fire events + features

Building-level data
(building area, occupancy)

Community-level statistical data
(socio-demographic factors)

MODEL

Training
fit · validate · select

Building-level generic fire-risk model

FIRE-RISK
ASSESSMENT

Fire events
(training output)

Building fire-risk
assessment

Aggregation

Community fire risk
(aggregated)

Sensitivity
analysis

DECISION

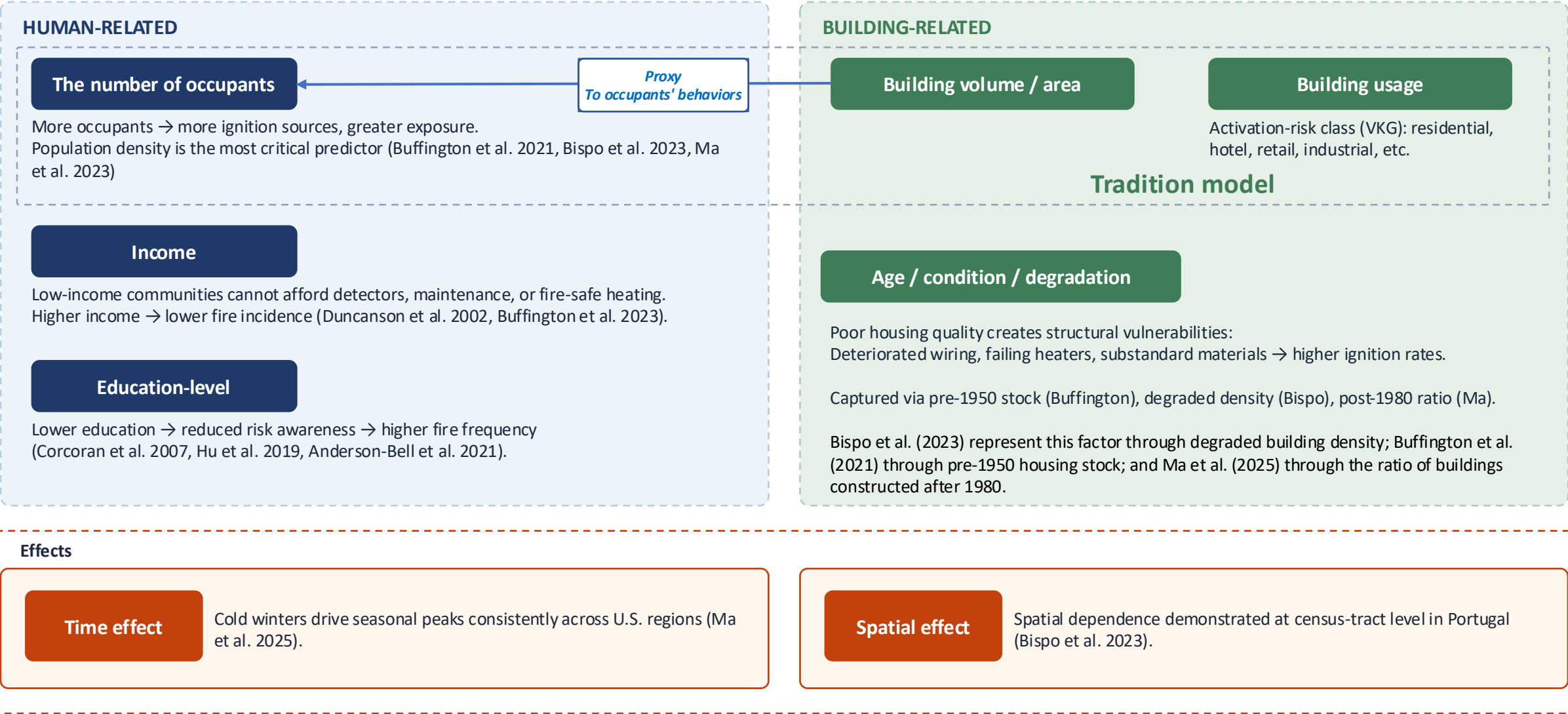
Building-level decisions
(risk-reduction measures)

Community-level decisions
(risk-reduction measures)

VOI

Critical Factors for Fire Occurrence

Predictors and effects identified in the empirical literature



Level 1 · Traditional (Volume-based) Models

Fire rate λ as a function of building volume / floor area and building (activation-risk) type

POISSON FRAMEWORK

Annual number of fire incidents:

$$N_F \sim \text{Poisson}(\lambda)$$

$$P(N_F = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

N_F = annual fire-incident count in a building

λ = fire rate (modelled at three levels)

Level 1: Volume

Level 2: Socio-demographic factors + Linear additive

Level 3: Nonlinear intercorrelated

LEVEL 1 · TRADITIONAL

Building volume / floor area as proxy for occupant population

$$\lambda = f(V, type)$$

V building volume

type building usage type / activation-risk class (VKG)

$f(\cdot)$ empirical fit (Barrois / Matrisk)

Limitation Ignores socio-demographic and temporal predictors.
Volume alone is a coarse proxy for the true ignition drivers.

Level 2 · Linear-Additive with Socio-Demographic Predictors

Volume + linear contributions from input variables, hierarchical across spatial scales

MODEL FORMULATION

Add a generalized linear-additive part:

$$\lambda = f(V) + X\beta$$

Assumption: Input variables are independent

X socio-demographic variables

β coefficient vector

Spatial-temporal extension (time effect):

$$\lambda = f(V) + X\beta + \text{time} \cdot \beta_t$$

Seasonal / temporal heterogeneity captured via β_t .

HIERARCHICAL BAYESIAN

Individual building → Community → City → Country

Country level

$$f(V), \mu_{coun}, \sigma_{coun}^2$$

City level

$$\mu_{city} \sim N(\mu_{coun}, \sigma_{coun}^2), \sigma_{city}^2$$

Community level

$$\beta \sim N(\mu_{city}, \sigma_{city}^2)$$

Assumption:

within the same community, β is the same at the building level

Level 3 · Nonlinear with Variable Correlations

Capturing interactions and complex relationships among input variables

VARIANT A · VOLUME INDEPENDENT

Building volume kept independent;
nonlinear function for the remaining input variables.

$$\lambda = f(V) + g(X)$$

$f(V)$ traditional volume-based contribution

$g(X)$ nonlinear function of socio-demographic / building factors

Captures complex correlations among inputs while preserving the well-understood role of building volume.

VARIANT B · FULLY GENERALIZED

All inputs, including building volume, enter a single nonlinear function.

$$\lambda = g(V, X)$$

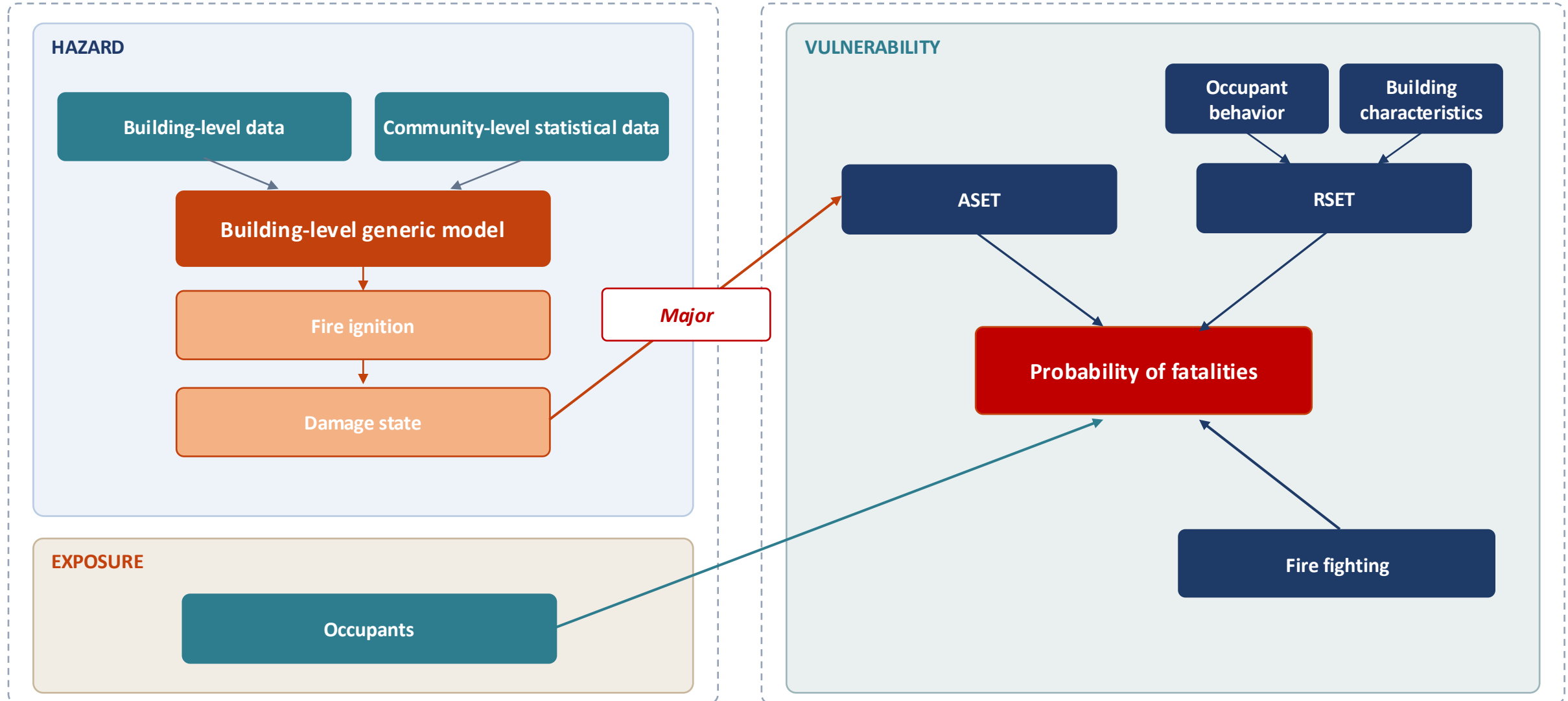
$g(\cdot)$ fully nonlinear (e.g. GAM, neural network)

Most flexible: captures interactions and correlations among all input variables.

Trade-off: less interpretability, higher data requirements.

Fire-Induced Fatalities

Hazard · Exposure · Vulnerability — performance-based fire-safety design for consequence analysis



ASET / RSET — Inputs and Sub-Timeline Uncertainty

Each sub-time is itself uncertain; ASET and RSET combine to determine fatality probability

ASET — Available Safe Egress Time

Definition

Time from ignition until conditions become untenable for occupants.

Drivers

Fire-growth curve (slow / medium / fast / ultra-fast), compartment geometry, ventilation, and the chosen untenability criterion.

Inputs

- Fire-growth rate
- Peak heat-release rate (HRR)
- Ventilation conditions
- Compartment volume

RSET — Required Safe Egress Time

Definition

Detection + alarm + pre-movement (reaction) + movement time required for occupants to evacuate.

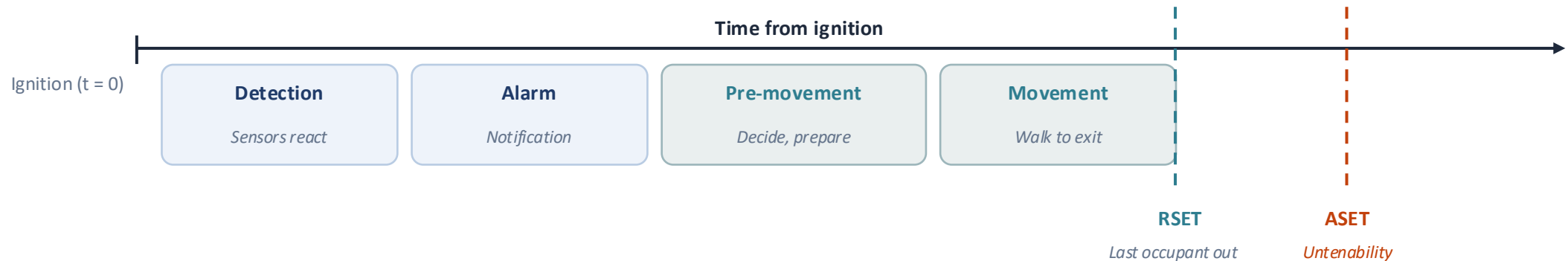
Drivers

Pre-movement varies from under a minute (alert workers) to many minutes (sleeping occupants, mobility-limited, or those attempting to fight the fire).

Movement depends on geometry, exit width, density, and walking speed.

Inputs

- Occupant population (age, mobility, familiarity)
- Pre-movement time distribution
- Exit availability and geometry
- Alarm reliability



Fatality occurs when RSET > ASET.

Sources of Uncertainty

Model and inputs for the model; ASET and RSET each contribute consequence-side uncertainty

Model

M

- **Parameters uncertainty**

uncertainty in fitted coefficients (β , σ^2) and prior choices

- **Model-structure uncertainty**

wrong functional form, omitted variables, residual errors

Input variables

X

- **Statistical-level inputs**

income, education, housing age reported at tract / census level — not available per individual building

Consequence analysis · ASET

A

- Ignition location & initial fuel load
- Heat-release rate · Fire-growth rate
- Flashover probability and timing
- Ventilation conditions (doors/windows open or closed?)
- Suppression activation time & effectiveness
- Zone-model uncertainty
- CFD model

Consequence analysis · RSET

R

- **Each sub-time is a random variable**

detection · alarm · pre-movement · movement

- **Population heterogeneity**

age, mobility, familiarity vary across the occupant pool

- **Behavioral uncertainty**

decisions under stress, mis-perceptions, panic, group dynamics

Conclusion

A generic probabilistic model connecting occurrence and consequence for building-level fire-risk analysis

Goal · A **generic probabilistic model** connecting **occurrence** and **consequence** for building-level fire-risk analysis.

Why probabilistic

Fire-risk analysis exposes uncertainty explicitly; instead of a conservative deterministic estimate, we report what we don't know and make better decision-making under risk with the distribution of possible outcomes.

Why building-level

Building-level models capture fine-grained building and occupant detail.

Why Bayesian networks

A graphical model encodes the conditional dependence structure among key factors; expert knowledge (and/or data) supplies the conditional probability tables; the DAG makes the inference logic intuitive.

Then what

Use the model to support decision-makers and policy-makers: reduce community-level fire risk and identify the most influential factors for each individual building.

Decisions

Risk-reduction and urban-planning decisions identified in the literature

RISK REDUCTION	Single building	URBAN PLANNING	Portfolio / community
● Intervention <i>Matellini et al. (2018) · Hao et al. (2023)</i>		● Resource allocation <i>Buffington et al. (2021) · Wu et al. (2026) · Song et al. (2017) · Hossain (2023)</i>	
● Performance-based structural fire design <i>Hasofer et al. (2007) · Chu & Sun (2008) · Gernay et al. (2016) · Zhang et al. (2017)</i>		● Fire-station location planning <i>Bispo et al. (2023) · Cui et al. (2024) · Lee et al. (2025)</i>	
		● Insurance risk stratification <i>De Sanctis & Fisher (2015) · Du et al. (2025)</i>	

Open questions

- Which decisions matter most for community fire-risk management?
- Given an estimate of fatality risk in a building, what decisions does it support, and for whom?
- In fire risk analysis/modeling, where does uncertainty matter most?

Data

Critical factors identified from literature and additional sources needed

FOR THE MODEL

Identified critical factors

Building information

Building volume
Building type
Building age / condition / degradation

Occupant information

Number of occupants
Income status
Education level

WANTED

Wanted data

Fire event reports

Building-level incident records.

Fire occurrence statistics

Census-tract / municipality fire-related data with socio-demographic predictors.

